



Genome Note

Complete genome sequence of an extensively drug-resistant *Pseudomonas aeruginosa* ST773 clinical isolate from North India

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ARTICLE INFO

Article history:

Received 20 May 2021

Revised 10 August 2021

Accepted 3 October 2021

Available online 26 October 2021

Editor: Prof Marco R Oggioni

Keywords:

Extensively drug-resistant

Pseudomonas aeruginosa

ST773

High-risk clone

Whole-genome sequencing

ABSTRACT

Objectives: *Pseudomonas aeruginosa* is an opportunistic Gram-negative pathogen causing a wide range of community- and hospital-acquired infections. Here we report the complete genome sequence of an extensively drug-resistant (XDR) *P. aeruginosa* strain (PA790) in order to understand the antibiotic resistance genes (ARGs) harboured by such a strain.

Methods: Whole-genome sequencing (WGS) was performed using Illumina HiSeq and Nanopore MinION platforms. Genome assembly was performed using Unicycler v0.4.8. The genome was annotated using the NCBI Prokaryotic Genome Annotation Pipeline (PGAP). In silico predictions were fulfilled using curated bioinformatics tools.

Results: *Pseudomonas aeruginosa* PA790 was classified as XDR and belongs to sequence type 773 (ST773). The complete genome size is 6 932 250 bp with a G+C content of 66.02% and a BUSCO (Benchmarking Universal Single-Copy Orthologs) score of 100. Strain PA790 harboured 12 different ARGs conferring resistance to eight different classes of antibiotics. It was identified as the nineteenth ST773 strain among 5785 whole-genome sequences of *P. aeruginosa* available in the NCBI database.

Conclusion: *Pseudomonas aeruginosa* PA790 belongs to ST773 and was identified as the nineteenth such isolate to be submitted to NCBI and the first complete ST773 genome from India. The WGS data with multiple ARGs of *P. aeruginosa* PA790 (ST773) will aid in understanding the evolution and phylogeny of such high-risk clones and provide a solid basis for further research on XDR strains.

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The emergence of extensively drug-resistant (XDR) *Pseudomonas aeruginosa* constitutes a major public-health concern worldwide [1]. *Pseudomonas aeruginosa* is an opportunistic pathogen exhibiting broad antimicrobial resistance that causes infections with high morbidity and mortality in hospitalised patients [2]. It is one of the six superbugs that pose a significant threat worldwide because of its capacity to become increasingly resistant to all available antibiotics [3,4]. The aim of this study was to determine the complete genome sequence of an XDR *P. aeruginosa* strain (PA790) and to perform a comparative analysis of all 19 genomes belonging to se-

quence type 773 (ST773) in order to understand their antimicrobial resistance profile.

Here we report the complete genome sequence of *P. aeruginosa* PA790 isolated from a patient with urinary tract infection at Sanjay Gandhi Postgraduate Institute of Medical Sciences (Lucknow, India) in January 2019. The isolate was identified using matrix-assisted laser desorption/ionisation time-of-flight mass spectrometry (MALDI-TOF/MS) (bioMérieux, Marcy l'Étoile, France). Antimicrobial susceptibility testing was performed by VITEK® 2 (bioMérieux) using an AST-N281 card according to the manufacturer's instructions. Isolate PA790 was identified as an XDR strain exhibiting resistance to amikacin (≥ 64 $\mu\text{g/mL}$), gentamicin (≥ 16 $\mu\text{g/mL}$), doripenem (≥ 8 $\mu\text{g/mL}$), imipenem (≥ 16 $\mu\text{g/mL}$), meropenem (≥ 16 $\mu\text{g/mL}$), ceftazidime (≥ 64 $\mu\text{g/mL}$), cefoperazone/sulbactam (≥ 64 $\mu\text{g/mL}$), cefepime (≥ 64 $\mu\text{g/mL}$), ciprofloxacin

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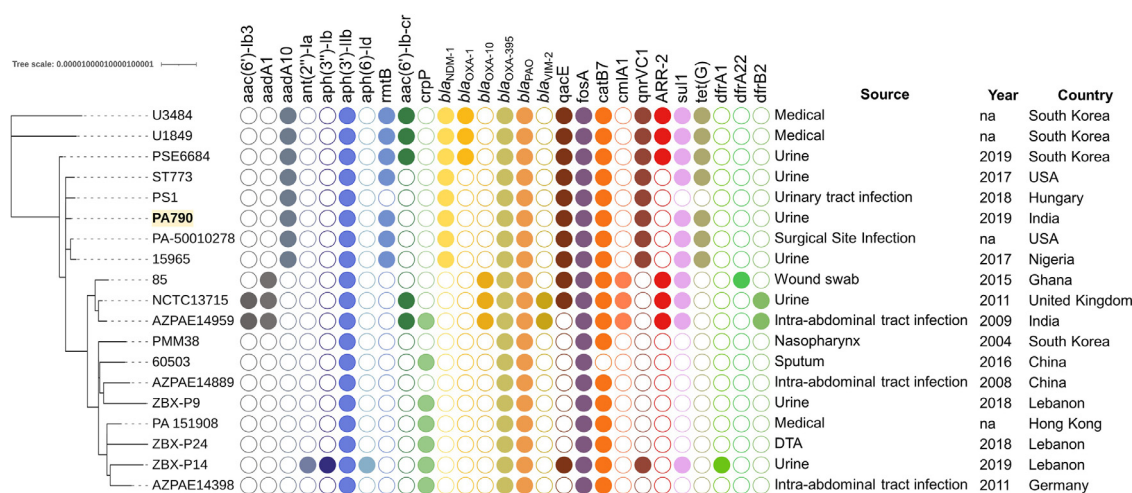


Fig. 1. Whole-genome phylogenetic tree and resistome analysis of 19 *Pseudomonas aeruginosa* genomes belonging to ST773. The phylogenetic tree was generated by the PhyML method using the REALPHY v.1.13 web server. Antibiotic resistance analysis was performed using ResFinder 4.1, and the presence/absence data were visualised with iTOL v.6. All 19 strains were isolated from human clinical samples. Details of source, year and country of isolation are shown. ST, sequence type; na, not available.

(≥ 4 $\mu\text{g/mL}$), levofloxacin (≥ 8 $\mu\text{g/mL}$), ticarcillin/clavulanic acid (≥ 128 $\mu\text{g/mL}$) and tigecycline (≥ 8 $\mu\text{g/mL}$) and intermediate resistance to colistin (≤ 0.5 $\mu\text{g/mL}$).

Pseudomonas aeruginosa PA790 was stored at -80°C and was subsequently grown overnight on a Mueller–Hinton agar plate at 37°C . A single colony was then picked and grown in Mueller–Hinton broth at 37°C for 16 h. Cells were collected by centrifugation and total genomic DNA was isolated using a DNeasy Blood and Tissue Kit (QIAGEN GmbH, Hilden, Germany) according to the manufacturer's protocol. The quality and quantity of extracted DNA were assessed by 0.8% agarose gel electrophoresis and NanoDrop 2000c UV-Vis spectrophotometer (Thermo Fisher Scientific, Wilmington, DE, USA), respectively.

Hybrid genome assembly of PA790 was generated using the reads from Illumina HiSeq X Ten sequencer (Illumina Inc., San Diego, CA, USA) and Oxford Nanopore MinION (Oxford Nanopore Technologies, Oxford, UK) platforms. FastQC was used to assess the sequencing quality of the reads. Quality-trimmed reads from Illumina (using Trim Galore v.0.6.5) and Nanopore (using Porechop v.0.2.4) were assembled using Unicycler v.0.4.8, and the final assembly quality was assessed with QUAST. The species of the assembled hybrid genome was confirmed as *P. aeruginosa* by ribosomal multilocus sequence typing (rMLST). MLST analysis using the PubMLST server and serotyping analysis using Past 1.0 revealed that strain PA790 belonged to ST773 and serogroup O4. It was identified as the nineteenth ST773 genome out of 5785 whole-genome sequences of *P. aeruginosa* available in the NCBI database. Genome completeness analysis with BUSCO v.4.1.2 using the 'bacteria_odb10' data set with 124 benchmarking universal single-copy orthologs (BUSCOs) showed the presence of 100% complete BUSCOs in the hybrid assembly. The complete genome assembly of PA790 is 6 932 250 bp with a mean G+C content of 66.02%. Whole-genome based phylogeny of the study genome (used as reference) along with the 18 other *P. aeruginosa* genomes belonging to ST773 was performed using the REALPHY 1.13 web server. Default parameters were used for all software unless otherwise specified.

The complete genome was annotated using the NCBI Prokaryotic Genome Annotation Pipeline (PGAP) v.5.2. Annotation revealed the presence of 6535 genes, of which 6383 were protein-coding sequences, 67 were tRNAs, 12 were rRNAs, 5 were ncRNAs and 68 were pseudogenes. Antibiotic resistance genes (ARGs) conferring resistance to aminoglycosides [*aadA10*, *aph(3'')-Ib*, *rmtB*], β -lactams [*bla*_{NDM-1}, *bla*_{OXA-395}, *bla*_{PAO}], disinfectant (*qacE*), fosfomycin (*fosA*),

phenicols (*catB7*), quinolones (*qnrVC1*), sulfonamides (*sul1*) and tetracycline (*tetG*) were identified using ResFinder 4.1. Comparative genome-wide antibiotic resistance analysis of PA790 with 18 other ST773 *P. aeruginosa* genomes revealed the presence of diverse ARGs in ST773 isolates (Fig. 1). According to the comparative analysis, 27 different ARGs conferring resistance to 12 different classes of antibiotics were identified among *P. aeruginosa* ST773 genomes.

In summary, the complete genome sequence of XDR *P. aeruginosa* PA790 analysed in this study offers insights into the antibiotic resistance profile of this high-risk ST773. The genome of ST773 *P. aeruginosa* PA790 is the first complete genome reported from India. This will aid in understanding the resistance patterns of such high-risk clones. Comparative resistome analysis suggested that ST773 is most likely to emerge as a global clone owing to the presence of diverse ARGs.

Nucleotide sequence accession numbers

The complete genome sequence of *P. aeruginosa* PA790 has been deposited in the NCBI GenBank database under accession no. [CP075176](#), BioProject [PRJNA729627](#) and BioSample [SAMN19136540](#).

Acknowledgments

SS and CPP acknowledge the Department of Science and Technology (DST/INSPIRE; Fellowship/2015/IF150708), Government of India, and the Indian Council of Medical Research (ICMR-SRF; AMR/Fellowship/8/2019-ECD-II) for fellowship support.

Funding

This study was supported by the Science and Engineering Research Board (SERB) [EMR/2015/001804], Government of India.

Competing interests

None declared.

Ethical approval

The study was approved by the institutional ethics committee of Sanjay Gandhi Postgraduate Institute of Medical Sciences, Lucknow, India [2015-123-EMP-87].

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